

Multi-scale finite elements and applications in fluid mechanics



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Context: very heterogeneous media

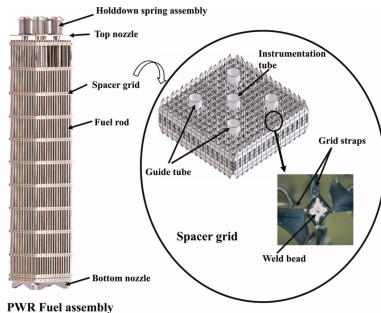
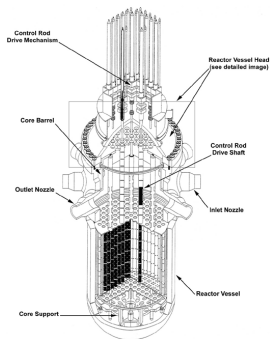
General idea of the Multi-scale finite elements

An abstract setting for the construction of such methods

Example in 1D; interesting consequences for standard FEs;
(new) Petrov-Galerkin methods for convection diffusion

Extension to multi-D with non-conforming finite elements for
fluid flow problems

Conclusions and perspectives

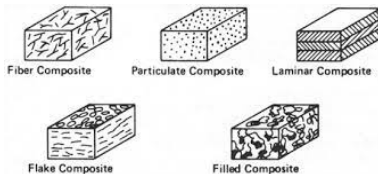


PWR Fuel assembly

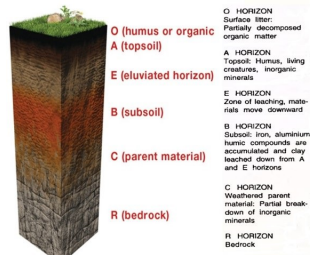
Nuclear reactor core: a multiscale medium with many obstacles.

Very expensive but still limited simulations on a partial, simplified assembly.

Is it possible to get an accurate CFD simulation (i.e. Navier-Stokes, possibly with turbulence models) of the entire core ?



Composite materials



Underground

Refrain from the idea that a coarse mesh will give usefull (although not very accurate) information:

On $D = (0; 1)$, $u_A(x) = x - x^2 + \mathcal{O}(\varepsilon)$ is the solution of

$$-\partial_x(A(x)\partial_x u) = 1 \quad , \quad A(x) = \frac{1}{2 + \cos(2\pi x/\varepsilon)}.$$

But if $H = n\varepsilon$ with $n \in \mathbf{N}^*$, then a P^1 Lagrange FE will not distinguish this from the solution of the same PDE with $\tilde{A}(x) = \frac{1}{\sqrt{3}}$ (because A and $\tilde{A}$ have the same integral on mesh segments), the solution of which is $u_{\tilde{A}}(x) = \frac{\sqrt{3}}{2}(x - x^2)$.

Example taken from Altmann, Henning, Peterseim, Acta Numerica 2021
(very good paper on numerical homogenization and multi-scale analysis)

Consider the variational formulation: look for $u \in H_0^1(\Omega)$ s.t.

$$a(u, w) = \ell(w) \quad \forall w \in H_0^1(\Omega) \quad \text{with} \quad a(u, w) = \int_{\Omega} A(x) \nabla u \cdot \nabla w.$$

Since $H_0^1(\Omega)$ has infinite dimension, we choose finite dimensional V_h and W_h (not necessarily included in $H_0^1(\Omega)$) and look for $u_h \in V_h$ s.t.

$$a_h(u_h, w_h) = \ell_h(w_h) \quad \forall w_h \in W_h.$$

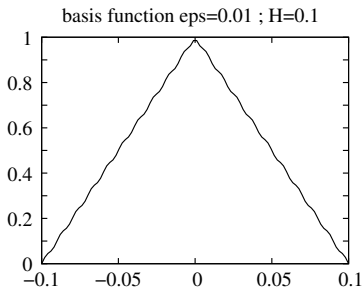
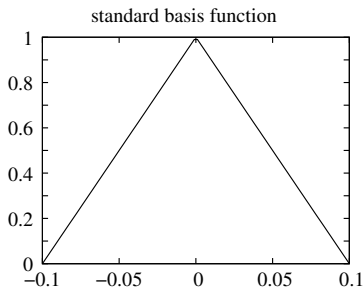
(Galerkin method if $W_h = V_h$, Petrov-Galerkin if not). And V_h approximates well $H_0^1(\Omega)$ when $h \rightarrow 0$.

Finite elements are special cases: defined on a mesh with piecewise polynomial, compactly supported basis functions.

But you can use something else!!! The idea is that basis functions solve the original PDE on each element of the mesh, with special boundary conditions (Efendiev and Hou, Multiscale finite element methods. Theory and applications. 2009).

Consider $-\partial_x(A(x)\partial_x u) = f$ with $A(x) = \frac{1}{2 + \cos(2\pi\frac{x}{\varepsilon})}$.

Basis functions solve $-\partial_x(A(x)\partial_x \phi_i) = 0$ with $\phi_i(x_j) = \delta_i^j$.



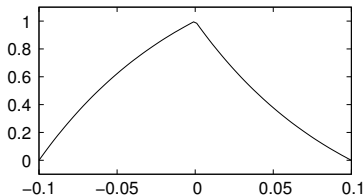
When $f \equiv 1$ and $\frac{H}{\varepsilon} \in \mathbb{N}$ using standard basis function underestimates the exact solution by a factor $\sqrt{3}/2$, while the modified method interpolates it exactly at the nodes (special case in 1D).

For the convection–diffusion equation $-\nu u'' + au' = f$,

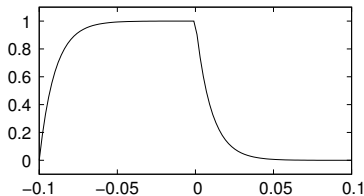
basis functions of V_H solve $-\nu \psi_i'' + a\psi_i' = 0$ with $\psi_i(x_j) = \delta_i^j$,

test functions of W_H solve $-\nu \phi_i'' - a\phi_i' = 0$ with $\phi_i(x_j) = \delta_i^j$.

basis function ($a/\nu = 10, h = 0,1$)



basis function ($a/\nu = 100, h = 0,1$)

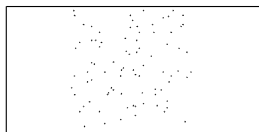


Coefficients of the associated matrix are

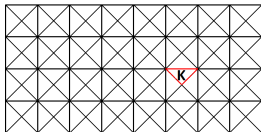
$$A_{i,i-1} = -\frac{a}{1 - \exp(-ah/\nu)}, \quad A_{i,i+1} = -\frac{a \exp(-ah/\nu)}{1 - \exp(-ah/\nu)}$$

$A_{i,i} = |A_{i,i-1}| + |A_{i,i+1}|$. No oscillations whatever the size of $\frac{ah}{\nu}$!

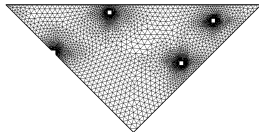
And if (f, ϕ_i) is computed exactly, we have $u_H(x_i) = u(x_i)$ (P.O. 2026)



Heterogeneous domain



Coarse mesh



Fine mesh

The MsFEM:

- ▶ (Petrov-)Galerkin method on the coarse mesh,
- ▶ Basis/Test functions **will not** be standard (polynomial) FE basis functions,
- ▶ They solve a certain PDE on each element of the coarse mesh with special boundary conditions,
- ▶ Approximations of these basis functions are computed on the fine mesh (e.g. by standard FEs).

Assume a mesh with N coarse cells; each made of n fine cells. Instead of a global problem with size $\mathcal{O}(N \times n)$, the multi-scale strategy has the following advantages:

- ▶ Meshing can be done in parallel, at least for non-conforming methods as will be discussed here,
- ▶ Solutions of $\mathcal{O}(N)$ local problems of size $\mathcal{O}(n)$ may be performed in parallel, since they are independent,
- ▶ Assembling the coarse problem can be done in parallel,
- ▶ Solution of coarse problem of size $\mathcal{O}(N)$ is inexpensive,
- ▶ Localised post-processing and visualisation of results is also without extra cost.

The Stokes problem with unknowns $\mathbf{u} \in H_0^1(\Omega)^d$, $p \in L_0^2(\Omega)$.

$$c((\mathbf{u}, p), (\mathbf{v}, q)) := \nu(\nabla \mathbf{u}, \nabla \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) - (\nabla \cdot \mathbf{u}, q) = (\mathbf{f}, \mathbf{v}).$$

Based on a coarse mesh with elements T and faces E , we set

$$V^{ext} = \left\{ \begin{array}{l} \mathbf{u} \in (L^2(\Omega^\varepsilon))^d \text{ s. t. } \mathbf{u}|_T \in (H^1(T \cap \Omega^\varepsilon))^d \forall T \\ \mathbf{u} = \mathbf{0} \text{ on } \partial\Omega^\varepsilon, \int_{E \cap \Omega^\varepsilon} [[\mathbf{u}]] \cdot \omega_{E,i} = 0 \forall E, j = 1, \dots, s \end{array} \right\},$$

The space V^{ext} extends the natural velocity space $(H_0^1(\Omega^\varepsilon))^d$ (the MsFEM is nonconforming). $\text{Span}(\omega_{E,i}) = (P_n(E))^d$.

The case $n = 0$ was treated by Muljadi, Narski, Lozinski, Degond (2015) (see also Le Bris, Legoll, Lozinski, 2012, 2014 for elliptic problems). We introduced and treated the theory for the case $n > 0$, (Feng, Allaire, P.O., 2022), and implemented $n = 1, 2$ (Balazi 2024, Balazi, Allaire, P.O. 2025).

The idea is to decompose $V^{ext} \times L_0^2$ into unresolved and resolved parts:

$$V^{ext} \times L_0^2 = (V_0 \times M_0) \oplus (V_H \times M_H)$$

- Definition of spaces of "unresolved" fields:

$$V_0 = \left\{ \mathbf{u} \in V_H^{ext} \mid \int_{E \cap \Omega^\varepsilon} \mathbf{u} \cdot \boldsymbol{\omega}_{E,i} = 0, \int_{T \cap \Omega^\varepsilon} \mathbf{u} \cdot \boldsymbol{\varphi}_{T,k} = 0 \right\},$$
$$M_0 = \left\{ p \in L_0^2 \mid \int_{T \cap \Omega^\varepsilon} p \boldsymbol{\varpi}_{T,j} = 0 \right\}.$$

The right combination is: $\text{Span}(\boldsymbol{\omega}_{E,i}) = (P_n(E))^d$,
 $\text{Span}(\boldsymbol{\varphi}_{T,k}) = (P_{n-1}(T))^d$ and $\text{Span}(\boldsymbol{\varpi}_{T,j}) = P_n(T)$.

Enriching only the set of edge weights $\boldsymbol{\omega}_{E,i}$ is insufficient: a given function $\mathbf{u}$ vanishing on the edges of any T would belong to the unresolved fine scales whatever the number of edge weights.

Since V_H^{ext} is not included in $(H_0^1(\Omega))^d$, we define the "broken" bilinear form c_H by

$$c_H((\mathbf{u}, p), (\mathbf{v}, q)) = \sum_{T \in \mathcal{T}_H} \int_{T \cap \Omega^\varepsilon} (\mu \nabla \mathbf{u} : \nabla \mathbf{v} - p \nabla \cdot \mathbf{v} - q \nabla \cdot \mathbf{u}).$$

Space $V_H \times M_H$ that is " c_H - supplementary" to $V_0 \times M_0$ i.e.
 $(\mathbf{u}, p) \in X_h \iff c_H((\mathbf{u}, p), (\mathbf{v}, q)) = 0, \forall (\mathbf{v}, q) \in V_0 \times M_0$:

$$M_H = \{q \in M \text{ s. t. } q|_T \in \mathbb{P}_n(T), \forall T \in \mathcal{T}_H\}, = \mathbb{P}_n^{\text{disc}}$$

$$V_H = \text{span} \{\Phi_{E,i}, \Psi_{T,k}\}$$

One basis function $\Phi_{E,i}$ associated to each edge E of the mesh and each weight $\omega_{E,i}$.

One basis function $\Psi_{T,k}$ associated to each element T of the mesh and each weight $\varphi_{T,k}$.

Basis function $\Phi_{E,i}$ associated to edge E and weight $\omega_{E,i}$:

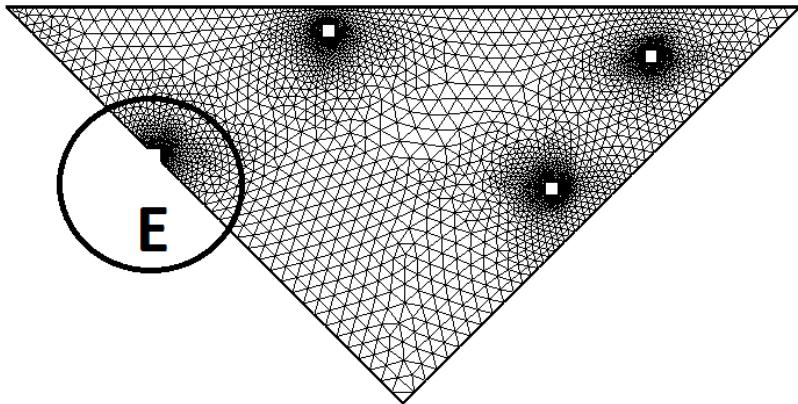
$$\left\{ \begin{array}{l} -\mu \Delta \Phi_{E,i} + \nabla \pi_{E,i} \in \text{span} \{ \varphi_{T_k,1}, \dots, \varphi_{T_k,r} \} \text{ in } T_k \cap \Omega^\varepsilon, \\ \text{div } \Phi_{E,i} \in \text{span} \{ \varpi_{T_k,1}, \dots, \varpi_{T_k,t} \} \text{ in } T_k \cap \Omega^\varepsilon, \\ \mu \nabla \Phi_{E,i} \mathbf{n} - \pi_{E,i} \mathbf{n} \in \text{span} \{ \omega_{F,1}, \dots, \omega_{F,s} \} \text{ on } F \cap \Omega^\varepsilon, \forall F \in \mathcal{E}(T_k), \\ \Phi_{E,i} = \mathbf{0} \text{ on } \partial B^\varepsilon \cap T_k, \\ \int_{F \cap \Omega^\varepsilon} \Phi_{E,i} \cdot \omega_{F,j} = \begin{cases} \delta_{ij}, & F = E \\ 0, & F \neq E \end{cases} \quad \forall F \in \mathcal{E}(T_k), \forall j = 1, \dots, s, \\ \int_{T_k \cap \Omega^\varepsilon} \Phi_{E,i} \cdot \varphi_{T_k,l} = 0 \quad \forall l = 1, \dots, r, \\ \int_{T_k \cap \Omega^\varepsilon} \pi_{E,i} \cdot \varpi_{T_k,m} = 0 \quad \forall m = 1, \dots, t. \end{array} \right.$$

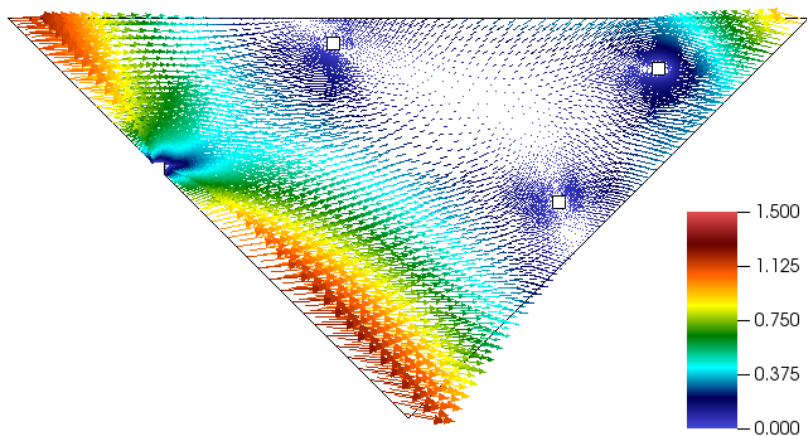
Basis function $\Psi_{T,k}$ associated to element T and weight $\varphi_{T,k}$:

$$\left\{ \begin{array}{l} -\mu \Delta \Psi_{T,k} + \nabla \pi_{T,k} \in \text{span} \{ \varphi_{T,1}, \dots, \varphi_{T,r} \} \text{ in } T \cap \Omega^\varepsilon, \\ \text{div } \Psi_{T,k} \in \text{span} \{ \varpi_{T,1}, \dots, \varpi_{T,t} \} \text{ in } T \cap \Omega^\varepsilon, \\ \mu \nabla \Psi_{T,k} \mathbf{n} - \pi_{T,k} \mathbf{n} \in \text{span} \{ \omega_{F,1}, \dots, \omega_{F,s} \} \text{ on } F \cap \Omega^\varepsilon, \forall F \in \mathcal{E}(T), \\ \Psi_{T,k} = 0 \text{ on } \partial B^\varepsilon \cap T, \\ \int_{F \cap \Omega^\varepsilon} \Psi_{T,k} \cdot \omega_{F,j} = 0 \forall F \in \mathcal{E}(T), \forall j = 1, \dots, s, \\ \int_{T \cap \Omega^\varepsilon} \Psi_{T,k} \cdot \varphi_{T,l} = \delta_{kl} \forall l = 1, \dots, r, \\ \int_{T \cap \Omega^\varepsilon} \pi_{T,k} \cdot \varpi_{T,m} = 0 \forall m = 1, \dots, t. \end{array} \right.$$

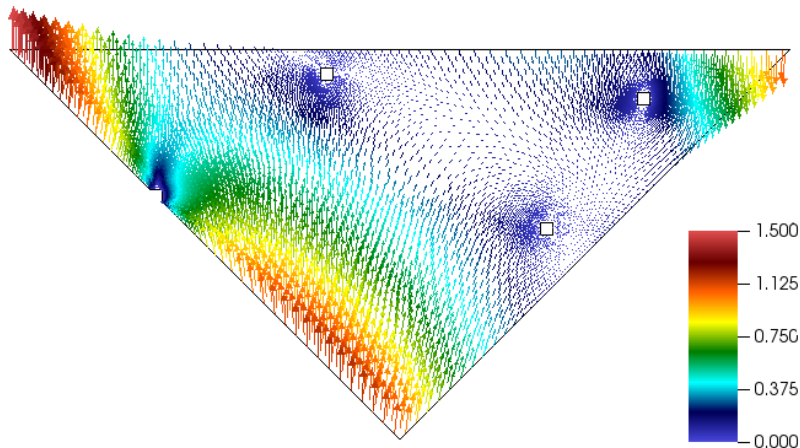
Because of the extra Lagrange multiplier, special high-order NC FEs are needed to solve these local problems (see Balazi, Allaire, Jolivet, P.O., NMPDE, 2025)

With obstacles in T :

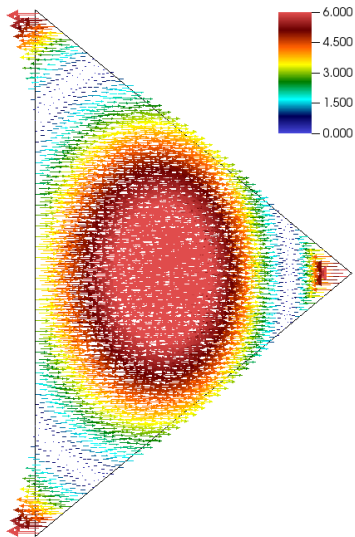




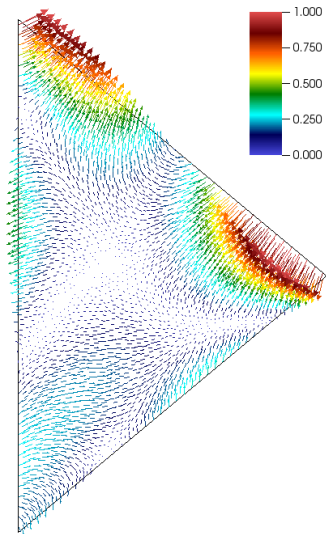
$$\int_E \mathbf{u}_{E,1} \cdot \mathbf{e}_1 = 1, \int_E \mathbf{u}_{E,1} \cdot \mathbf{e}_2 = 0, \int_{E'} \mathbf{u}_{E,1} \cdot \mathbf{e}_j = 0.$$

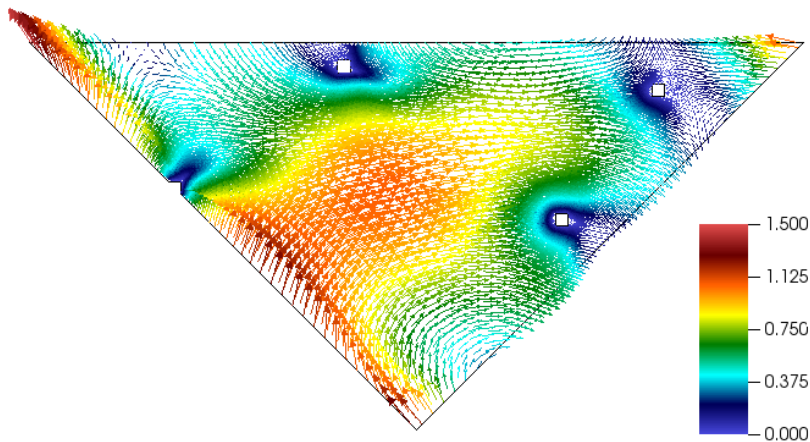


$$\int_E \mathbf{u}_{E,2} \cdot \mathbf{e}_2 = 1, \int_E \mathbf{u}_{E,2} \cdot \mathbf{e}_1 = 0, \int_{E'} \mathbf{u}_{E,2} \cdot \mathbf{e}_j = 0.$$



$$\int_E \mathbf{u} \cdot \boldsymbol{\omega}_{E,i} = 0, \quad \int_T \mathbf{u} \cdot \mathbf{e}_1 = 1, \quad \int_T \mathbf{u} \cdot \mathbf{e}_2 = 0.$$





In the case of a periodic configuration, using homogenisation results, we may prove that (ε being the size of the obstacles and $H \gg \varepsilon$)

$$\begin{aligned} & \varepsilon^{-1} |\mathbf{u}_\varepsilon - \mathbf{u}_H|_{H,1} + \|p_\varepsilon - p_H\|_{L^2(\Omega^\varepsilon)} \\ & \leq C \left[H^n \|\mathbf{f} - \nabla p^*\|_{H^n(\Omega)} + H^n |p^*|_{H^{n+1}(\Omega)} \right. \\ & \quad \left. + \left(\sqrt{\varepsilon} + \sqrt{\frac{\varepsilon}{H}} \right) \|\mathbf{f} - \nabla p^*\|_{H^2(\Omega) \cap C^{1,\alpha}(\overline{\Omega})} \right] \end{aligned} \quad (1)$$

where the constant C does not depend on H , ε and $\mathbf{f}$, while p^* is the homogenized pressure, solution of a Darcy equation

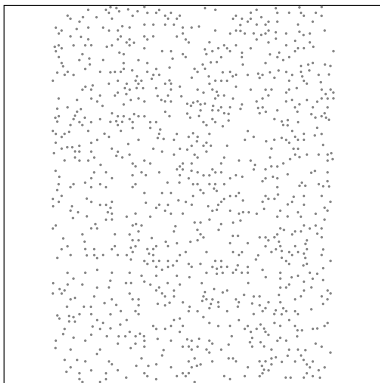
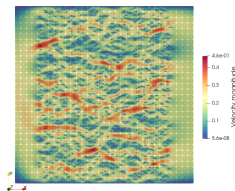
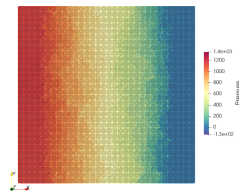


Figure: Domain with 1024 randomly located obstacles.

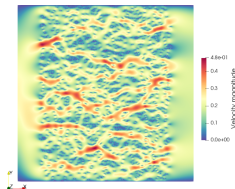


(a) Velocity magnitude

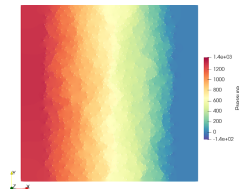


(b) Pressure field

Figure: MsFEM approximation, $n = 2$, (4096 coarse elements)



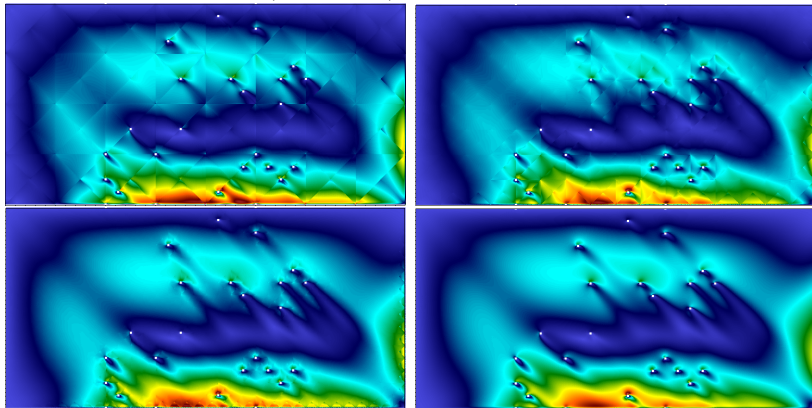
(a) Velocity magnitude



(b) Pressure Field

Figure: Reference solution

Oseen flow with $U_0 = (400, -400)$ and 26 obstacles of size 1.5×10^{-2}



$n = 1$, from left to right, top to bottom: $4 \times 8 \times 4$ coarse triangles, $8 \times 16 \times 4$ triangles, $16 \times 32 \times 4$ triangles, reference on a fine mesh with 2 million triangles.

- ▶ Construction of Multi-Scale Finite Elements
- ▶ Provides new insight on Standard Finite Elements
- ▶ Can reproduce complex fluid flows with low cost
- ▶ More tests with Oseen and Navier-Stokes flows
- ▶ A priori and a posteriori analysis including effects of approximate resolution of fine-scale basis functions
- ▶ Non-stationary problems
- ▶ Non-linear problems: turbulence modelling
- ▶ Links with Domain Decomposition